Ahmedabad University

Algorithmic Portfolio Optimization

Milestone - 2

Problem Motivation

- Markets are inherently stochastic systems
- Price movements unpredictable
- Deterministic models fail
- Probabilistic models capture uncertainty
- Investors must allocate capital rationally
- Goal: balance return and risk mathematically
- Requires formal modeling approach

Mathematical Problem Statement

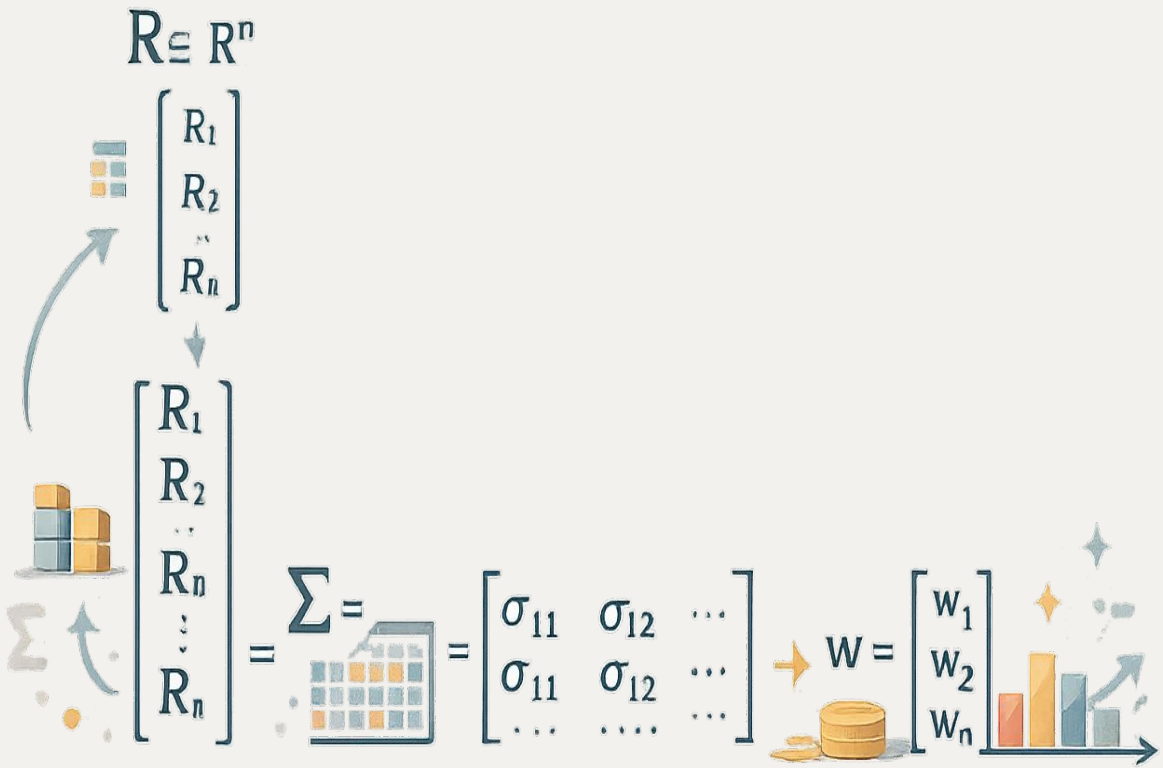
- Given n assets
- Return vector R = (R1…Rn)
- Weight vector w = (w1…wn)
- Find optimal w
- Maximize objective function f(w)
- Subject to constraints
- Optimization domain is feasible region

Random Variable Modeling

- Each asset return Ri is random variable
- Ri ∈ ℝ continuous
- Distribution estimated from data
- Mean μi = E[Ri]
- Variance σ^2i = Var(Ri)
- Captures uncertainty quantitatively
- Forms probabilistic foundation

Vector Representation

- Return vector $R \in \mathbb{R}^n$
- Weight vector $w \in \mathbb{R}^n$
- Mean vector $\mu \in \mathbb{R}^n$
- Covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$
- Compact mathematical notation
- Enables matrix operations
- Simplifies computation



Portfolio Return Formula

- $R_p = \Sigma\, w_i R_i$
- Matrix form: $R_p = w^T R$
- Linear combination property
- Distribution depends on w
- Different weights → different distribution
- Portfolio design = distribution shaping
- Key decision variable = w

Expected Return

- $E[R_p] = \Sigma\, w_i \mu_i$
- Vector form: $w^T \mu$
- Linearity of expectation
- Independent of covariance
- Measures average performance
- Higher expectation desirable
- First optimization objective

Risk Definition

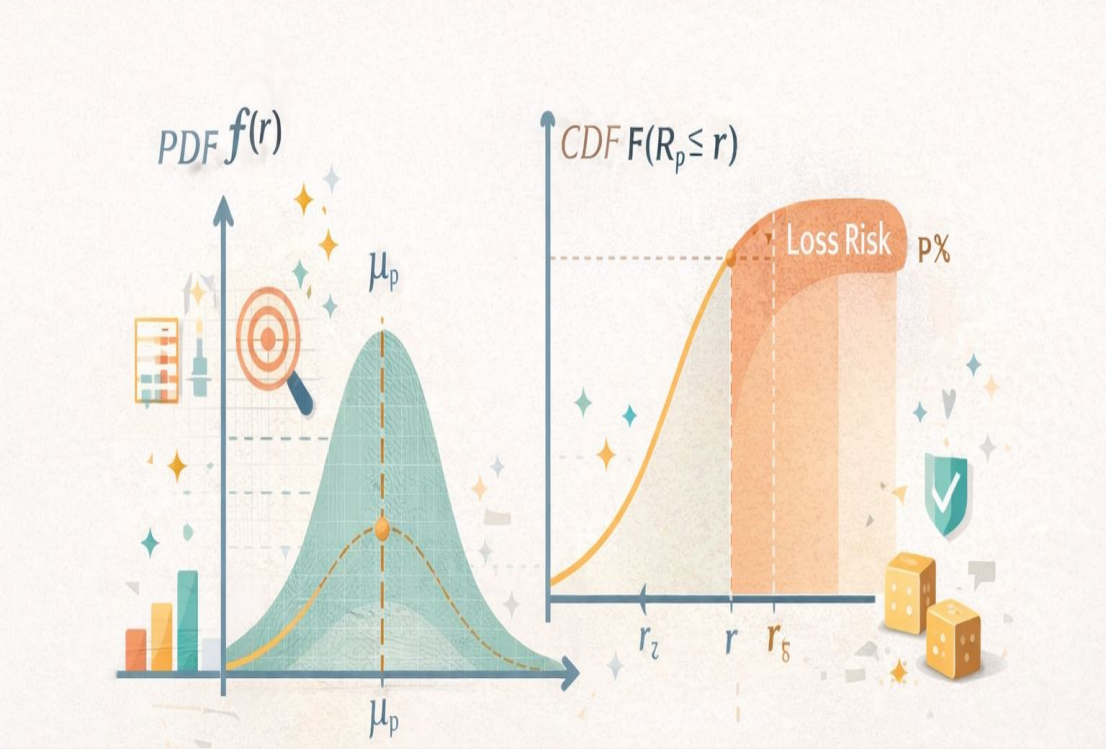
- Risk measured via variance
- $\sigma^2 p = Var(Rp)$
- $\sigma^2 p = w^T \Sigma w$
- Quadratic function of weights
- Captures uncertainty magnitude
- Higher variance = higher risk
- Second optimization objective

Covariance Matrix Role

- $\Sigma ij = Cov(Ri,Rj)$
- Diagonal = individual variances
- Off-diagonal = dependencies
- Negative covariance reduces risk
- Positive covariance increases risk
- Captures diversification effect
- Most critical risk component

Probability Interpretation

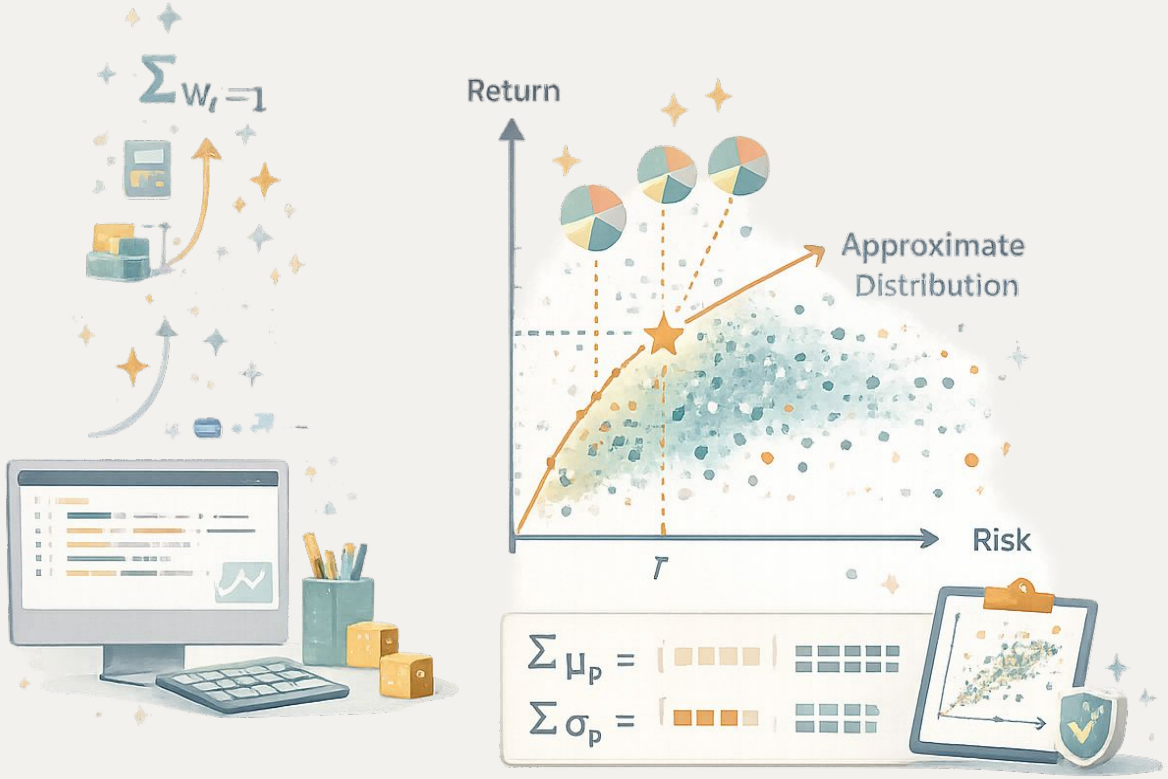
- Rp is random variable
- PDF f(r) shows likelihood
- CDF F(r)=P(Rp≤r)
- Tail probability = loss risk
- Distribution shape matters
- Not just mean important
- Risk assessment via probability



Monte Carlo Simulation

- Generate random weight vectors
- Constraint Σwi=1
- Simulate thousands portfolios
- Compute return & risk each
- Approximate true distribution
- Law of large numbers applies

Numerical probability estimation



Optimization Objective

- Sharpe Ratio S = (E[Rp] – rf)/σp
- Measures risk-adjusted return
- Higher S better
- Optimization goal: maximize S
- Equivalent: minimize –S
- Scalar objective required

Defines optimality mathematically

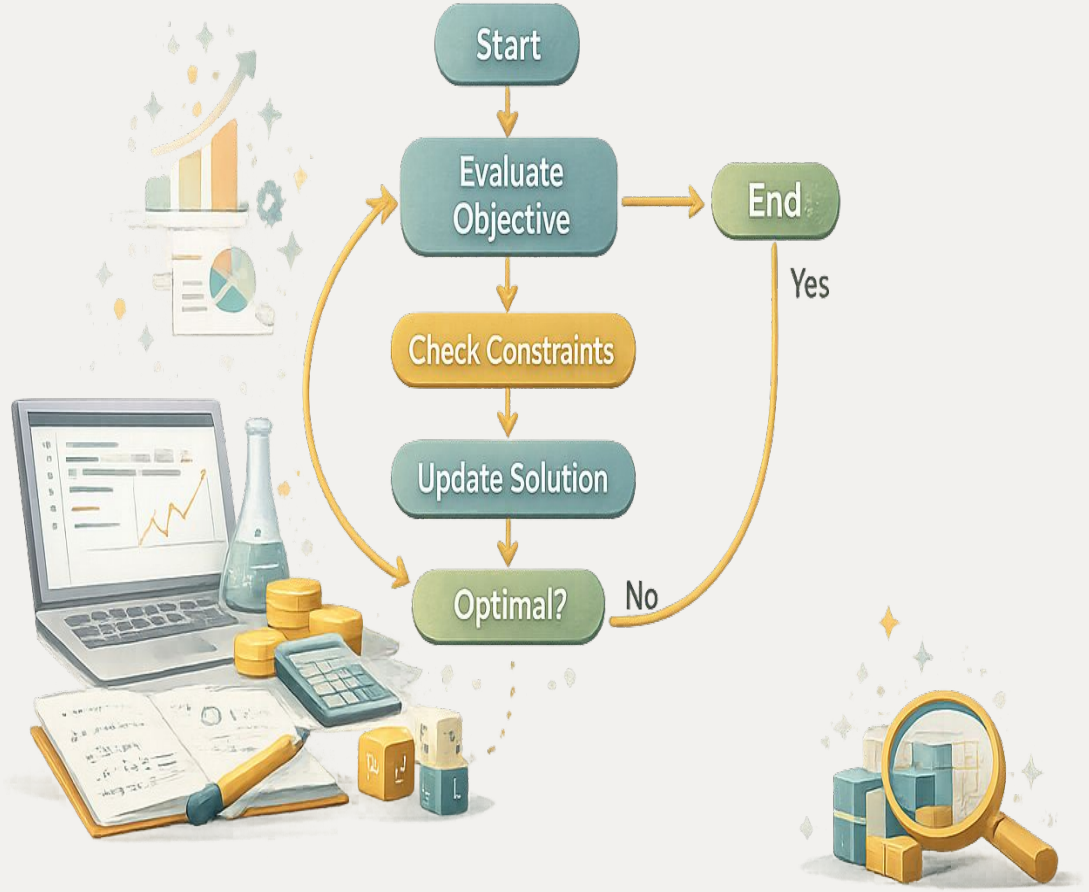
Constraint System

- Budget constraint $\Sigma w_i = 1$
- Non-negativity $w_i \geq 0$
- Upper bounds $w_i \leq 1$
- Feasible region convex
- Ensures realistic portfolios
- Prevents leverage,

defines search space

Optimization Algorithm

- Method: SLSQP
- Sequential quadratic programming
- Handles nonlinear objectives
- Handles equality constraints
- Handles inequality constraints
- Uses gradients, iteratively converges



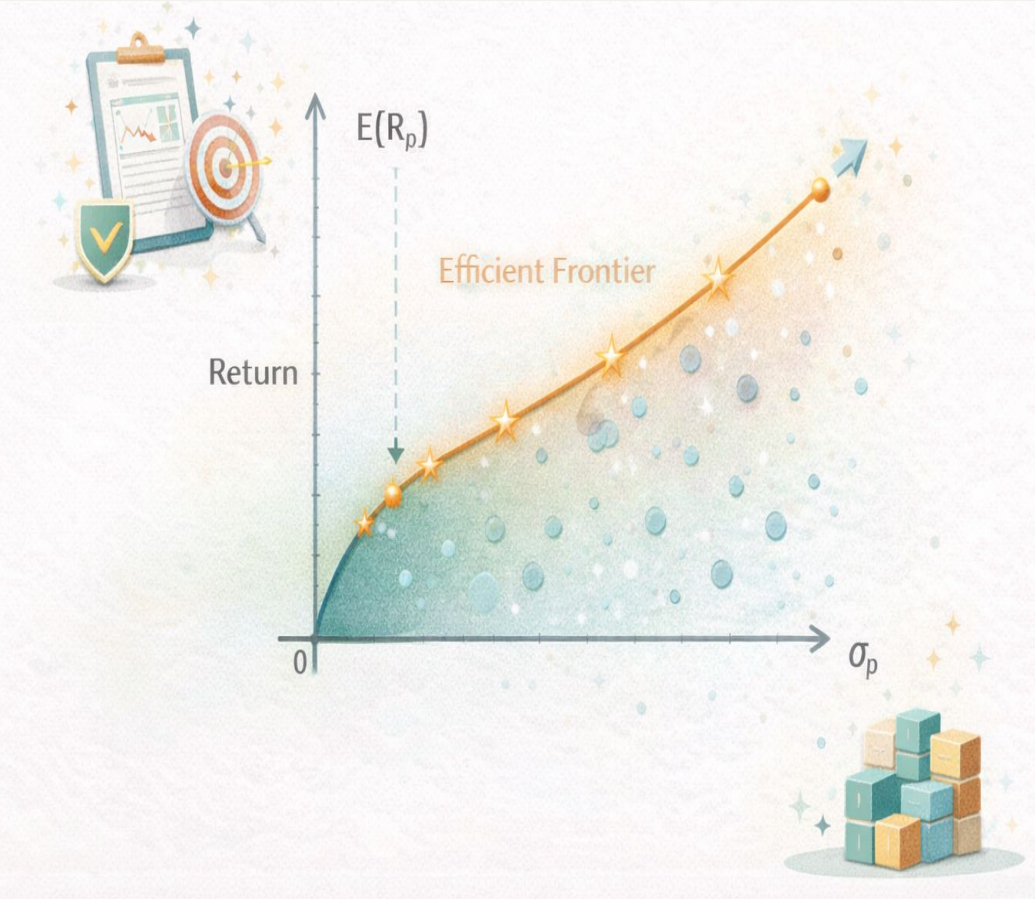
Quadratic Nature of Problem

- Variance term $w^T \Sigma w$ quadratic
- Objective nonlinear
- Convex region often
- Single global minimum possible
- Efficient numerical solving
- Mathematical structure advantageous
- Optimization stable

Efficient Frontier

- Set of optimal portfolios
- Solve min σp for fixed return
- Parametric optimization
- Forms curve in risk-return plane
- Each point efficient
- Below frontier inefficient

Represents best trade-offs



Capital Market Line

- Line tangent to frontier
- Equation $E[R]=rf+S\sigma$
- Slope = Sharpe ratio
- Tangency portfolio optimal
- Combines risk-free + risky assets
- Defines market equilibrium

Optimal investor choice

Statistical Assumptions

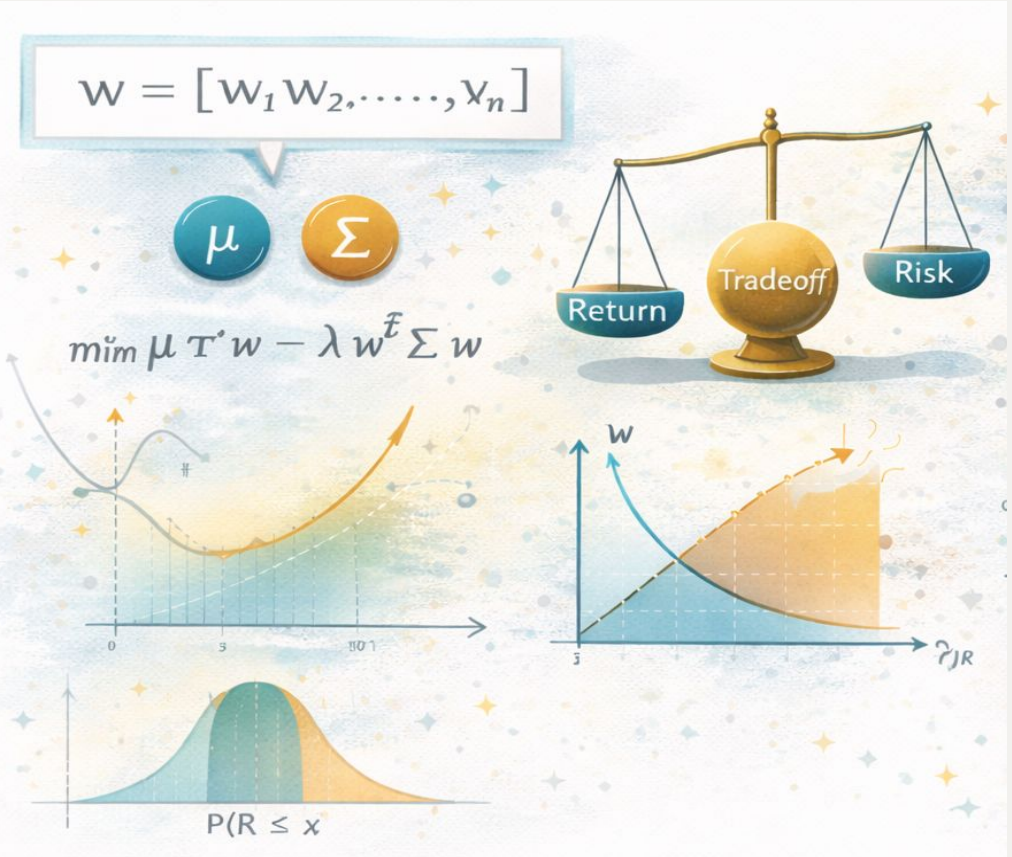
- Returns stationary
- Covariance constant
- Distributions stable
- Historical data valid
- No transaction cost
- No liquidity constraint, simplifies modeling

Limitations

- Real markets dynamic
- Correlations change
- Extreme events rare
- Model sensitive to inputs
- Estimation error exists
- Optimization unstable sometimes requires interpretation

Final Mathematical Insight

- Optimization variable = weight vector
- Objective depends on μ and Σ
- Tradeoff between return and risk
- Quadratic programming problem
- Solution gives optimal allocation
- Mathematics guides decisions
- Probability quantifies uncertainty



Conclusion

- Portfolio optimization is decision theory
- Randomness central element
- Probability models uncertainty
- Linear algebra enables computation
- Optimization finds best solution
- Framework generalizable, applicable beyond finance